

Two-Step Equations

AQA · KS3 Year 8

Answer each question in your workbook. Show your working.

Method

Spot the letter term and the + / - number Undo the + or - first (both sides) Now it is a one-step equation

Undo the $\times$ or $\div$ (both sides) Check by substituting

Common mistakes

- Dividing before undoing the + or -.
- Not doing each step to both sides.
- $2x = 8$ gives $x = 4$, not $2x = 4$.
- Losing a minus sign with negative constants.

Worked example

Solve $2x + 3 = 11$.

Subtract 3: $2x = 8$. Divide by 2: $x = 4$.

Check: $2 \times 4 + 3 = 11$

Visual aid

Visual Example

$$2x + 3 = 11$$

$$\text{-3 both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



Balance scale: $2x + 3$ equals 11. Remove 3 from both sides to get $2x = 8$, then halve both sides to get $x = 4$.

Questions

Answer each question in your workbook. Show your working.

Q1 of 8

[2] ● ○ ○ ○ ○

Solve: a) $2x + 1 = 7$ b) $3x + 4 = 19$

Q2 of 8

[2] ● ○ ○ ○ ○

Solve: a) $5x - 2 = 13$ b) $4x - 7 = 9$

Q3 of 8

[2] ● ● ○ ○ ○

Solve: a) $x \div 2 + 3 = 8$ b) $x \div 3 - 1 = 4$

Q4 of 8

[3] ● ● ○ ○ ○

Solve: a) $2x + 9 = 3$ b) $5x + 14 = 4$ c) $3x - 5 = -11$

Q5 of 8

[3] ● ● ● ○ ○

Solve: a) $7 = 2x + 1$ b) $10 = 4x - 6$ c) $(x + 5) \div 2 = 6$

Q6 of 8

[3] ● ● ● ○ ○

Form and solve: a) Taxi £3 + £2/mile = £17 — how many miles? b) Number $\times 3$ then $- 5 = 16$.

Q7 of 8

[3] ● ● ● ● ○

Rectangle length $(2x+1)$, width 4, perimeter 30 cm. a) Equation. b) Find x . c) Length.

Q8 of 8

[5] ● ● ● ● ●

Repair: £15 call-out + £c/hour. 3-hour job = £60. a) Find c . b) Cost of 5 hours. c) £75 job — how many hours?